ANSWER KEY & MARKING SCHEME SET 09

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** citric acid	oranges, lemon
2	**(c)** neutral	neutralisation
3	**(d)** filament	glows white hot
4	**(c)** Plastic is an insulator and protects from shock	safety
5	**(c)** diamond	a form of carbon
6	**(c)** sonority	e.g. bells
7	**(d)** carbon dioxide	turns lime water milky
8	**(c)** a reversible physical change	salt is recovered
9	**(a)** the body is growing rapidly	growth spurt
10	**(c)** seek help from parents or teachers	handle wisely
11	**(c)** conduction	contact
12	**(d)** poor conductors of heat	do not burn hand
13	**(d)** at night or on cloudy days	needs sunlight
14	**(a)** 50 km/h	speed = distance ÷ time
15	**(c)** Liver	stored in gall bladder
16	**(c)** circulatory system	heart and blood
17	**(b)** oxygen	just like animals
18	**(c)** glucose	later stored as starch
19	**(d)** 45° to the tube	parallel to each other
20	**(c)** leap year	366 days

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	Acids taste sour and turn blue litmus red (1); bases taste bitter and feel soapy (1).	1+1
22	A closed path along which electric current flows (2).	2
23	Painting; galvanisation; oiling (2).	2
24	Isolation and impurity are myths (1); it is a natural process (1).	1+1
25	Transfer of heat without a medium (1); through space as heat waves (1).	1+1
26	Second (1); metre per second (1).	1+1
27	Breakdown of food into simpler forms (1); mouth, oesophagus, stomach, small intestine, large intestine, anus (1).	1+1
28	Traps sunlight energy for photosynthesis (2).	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Lemon juice contains dissolved substances; pure water has none (3).	3
30	Fuels with a low value catch fire easily (1.5); kerosene more easily than paper (1.5).	1½+1½
31	Measure body temperature (1.5); shake before use, wash after use (1.5).	1½+1½
32	Thin walls with blood vessels (3).	3
33	Orientation changes the shadow shape (1.5); e.g. coin edge-on looks line (1.5).	1½+1½
34	Day and night are about equal (3).	3
35	Changes soil acidity (1.5); lowers fertility and harms plants (1.5).	1½+1½
	OR: Lime or slaked lime (1); it is a base and neutralises the soil acidity (2).	1+2

36	Pots A and B in sunlight, one watered (1.5); watered grows better (1.5).	1½+1½
	OR: No gas exchange or transpiration (1.5); photosynthesis stops (1.5).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) Keep air and water away (1). (ii) Painting, oiling, galvanising (3). (iii) Saves money and resources (1).	1+3+1
	OR: (i) Basic and acidic (2). (ii) Litmus test (2). (iii) MgO; SO_2 (1).	2+2+1
38	(i) Safe and respectful interaction (2). (ii) Use for learning, protect privacy (2). (iii) Think before posting; avoid strangers (1).	2+2+1
	OR: (i) Diet, hygiene, exercise, sleep, social balance (3). (ii) Explanation (2).	3+2
39	(i) A (1). (ii) 72 km and 54 km (2). (iii) A (1). (iv) 18 km (1).	1+2+1+1
	OR: (i) No (1). (ii) 5 m/s (1). (iii) 6 m/s (1). (iv) 5.33 m/s (2).	1+1+1+2
40	(i) Mirror (1). (ii) Smooth surface (1). (iii) Regular and diffused (1). (iv) Still water, rough road (2).	1+1+1+2
	OR: (i) 4 m (2). (ii) 2 m (2). (iii) Image is as far behind as object is in front (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND